

MTPA Guide

Power Application Controller™



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1 DESCRIPTION

For a salient PMSM, d-axis and q-axis inductances are different, which needs to generate the optimized d-axis and q-axis current reference to optimize the torque generation. It is commonly called Maximum Torque per Ampere (MTPA) control.

QORVO FOC firmware includes the feature of MTPA for a salient PMSM. Before enabling MTPA feature, please refer to auto-tuning guide (..\resources\guides\features) to get the correct motor parameters including motor resistance, inductance and back-EMF constant.

2 FIRMWARE FLOW

The MPA algorithm will automatically calculate the optimized MPA angle as illustrated in Figure 1. With a motor having different L_d and L_q , MPA angle will be updated from the given motor parameters as shown in Figure 2. Typically, the MPA angle is zero for a non-salient PMSM.

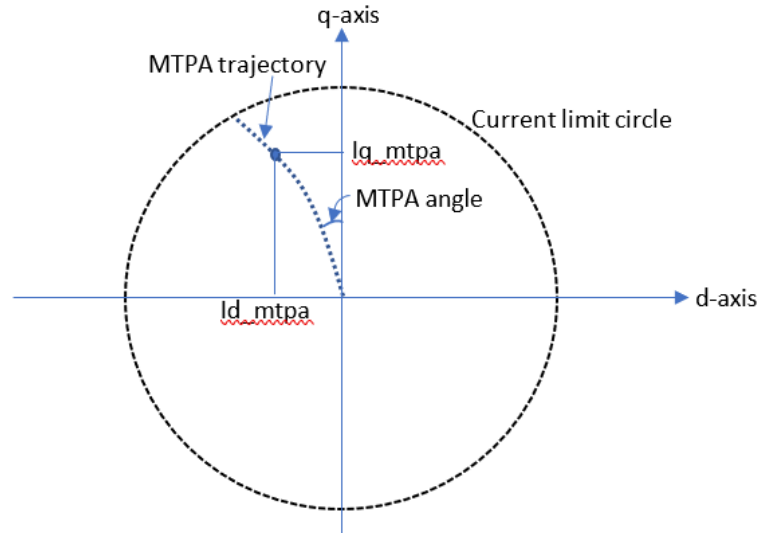


Figure 1: Control trajectory for MPA

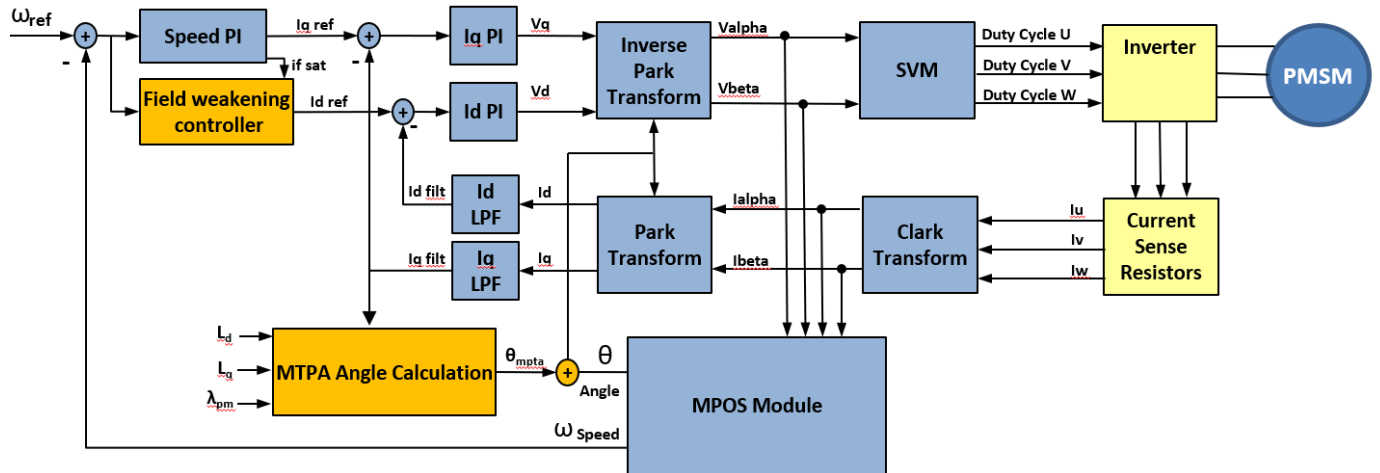


Figure 2: FOC control diagram with MTPA block

3 USING MTPA FEATURE

3.1 Enabling MTPA

To enable the MTPA feature, find the compile option in config_features.h and verify that it is defined:

```
#define ENABLE_MTPA
```

3.2 Firmware Options

To configure MTPA, please verify that the desired options are defined in config_features.h and that the selections and values are appropriate for the application:

MTPA Configuration

```
I_FILTER_CUTOFF_FREQ_Q16
```

This applies a low pass filter on the measured motor current for generating MTPA angle.

```
MTPA_RAMP_RATE_Q16
```

This enables the ramp control to reach the desired MTPA angle during transient conditions.

```
MTPA_ANGLE_AUTO_CALC
```

This enables the automatic calculation of the MTPA angle based on the given motor parameters, such as R, Ld, Lq and Ke.

3.3 MTPA angle Display

MTPA angle can be exported in Parameter group 3 as below for display purposes. The debug feature needs to be enabled in config_debug.h:

```
case 99:
    parameter_value1 = motor_ptr->mtpa.mtpa_coff_q16[1];
    parameter_value2 = motor_ptr->mtpa.mtpa_coff_q16[0];
    parameter_value3 = motor_ptr->mtpa.mtpa_angle_rad_q16;
```

3.4 Tuning Parameters (GUI)

There are no GUI configurable parameters for MTPA.



CONTACT INFORMATION

For the latest specifications, additional product information, worldwide sales and distribution locations:

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